

Alisina Karimi

Turin, Italy | alisinakarimi.2003@gmail.com
[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

Summary

Software Engineer specializing in scalable systems and distributed architectures. B.Sc. in Computer Engineering with a track record of optimizing production systems, reducing infrastructure costs via architectural improvements, and building end-to-end analytics platforms. Experienced in high-autonomy remote environments across multiple time zones.

Experience

WineHub Commerce

Nov 2024 – Present

Full Stack / Backend Engineer

Adelaide, Australia (Remote)

- Optimized database queries and performance, achieving **3×–10× improvements** across critical workloads.
- Worked on migration from MariaDB to **PostgreSQL**, resulting in a **60% infrastructure cost reduction**.
- Designed and implemented **Go-based microservices** for analytics, billing, and webhook processing.
- Built a full BI platform (custom dashboards, reporting pipelines) for financial decision-making.
- Integrated compliance/fulfillment APIs: Sovos ShipCompliant, Avalara, VinoShipper, and WineDirect.
- Automated reporting and compliance pipelines, significantly reducing operational overhead.

Moneyviz srl

May 2024 – Nov 2024

Full Stack Developer

Turin, Italy

- Developed financial analytics features using **Laravel** and **Vue.js**, enabling portfolio performance tracking and visualization for end users.
- Improved backend data ingestion and transformation pipelines, reducing latency of analytical queries and improving responsiveness of dashboards.
- Refactored data processing logic to improve consistency across financial reporting workflows and reduce edge-case discrepancies.
- Collaborated closely with product stakeholders to translate financial requirements into scalable backend and UI implementations.

RUN Polito APS-ETS

Sep 2022 – May 2024

Software Developer

Turin, Italy

- Developed internal tools using **Go** and **JavaScript** to support operational workflows for a multi-team student organization.
- Automated administrative and coordination processes, reducing manual workload and improving turnaround time for internal requests.
- Refactored legacy services to improve maintainability and reliability, reducing recurring issues and simplifying future feature development.
- Worked directly with non-technical stakeholders to identify inefficiencies and design practical software solutions.

WEEEOpen

May 2022 – Aug 2022

Back-end Developer

Turin, Italy

- Improved backend services using **Django**, enhancing performance and stability of core application features.
- Supported frontend integration using **React**, improving data flow between backend APIs and user interfaces.
- Participated in debugging and issue resolution across the stack, reducing friction in development and testing cycles.

Open Source / Freelance Developer

Sep 2021 – Present

Software Developer

Remote

- Designed and implemented distributed data processing prototypes using **Kafka** and **Node.js** for event-driven workflows.
- Contributed to open-source backend projects, focusing on API design, documentation, and developer experience improvements.
- Worked independently on multiple small-scale projects, strengthening ability to scope, design, and deliver features end-to-end.

Projects

Chatster | *Go, React, SQLite, Gorilla WebSocket, Prometheus, Docker, GitHub Actions*

- Designed a **real-time chat** reference stack: WebSocket hub with buffered broadcast, **per-client write serialization** for safe concurrent use of Gorilla WebSocket, and **SQLite**-backed history with configurable replay of recent messages on connect.
- Shipped operator-facing endpoints: structured **JSON logging** (slog), **graceful shutdown**, /health with database ping (degraded 503), and /metrics for **Prometheus**.
- Added **abuse controls**: maximum username/message sizes, **per-IP WebSocket upgrade rate limiting**, and optional **Origin** allowlist for production hardening.
- Automated quality with **GitHub Actions** (Go tests with race detector, coverage, **ESLint** on the React SPA), **Docker Compose** for local full-stack runs, and portfolio-grade architecture, threat-model, and scaling documentation.

Quorabust | *Python, XGBoost, scikit-learn, HTTP API, Prometheus, Docker, pytest*

- Built an installable **Python** library for **Quora-style duplicate question detection**: text cleaning, pairwise **TF-IDF** and lexical features, **XGBoost** classifier, and optional **embedding**-based feature backends.
- Delivered **train and serve CLIs** plus an HTTP service exposing prediction, **Prometheus** metrics at /metrics, interactive **API documentation**, and optional **A/B** model selection via request headers for controlled rollouts.

- Persisted models with rich **metadata**: training counts, evaluation metrics, dataset checksums, and reference feature statistics to support **reproducibility** and **drift** awareness when loading trusted artifacts only.
- Enforced engineering rigor with **pytest** and coverage thresholds, **Ruff** and **mypy**, **pre-commit**, container-friendly packaging, and operational guides for **load testing** and **Grafana** monitoring.

news-api | *TypeScript, Express, Node.js, Redis, OpenAPI, Docker, Kubernetes*

- Implemented a **production-oriented** news search API on **Express** and **TypeScript** integrating the GNews provider with a **pluggable cache** (in-memory or **Redis**) to deduplicate identical queries and protect upstream quota.
- Hardened the edge: **Helmet** security headers, configurable **rate limiting**, optional **client API keys** on /api/*, **TRUST_PROXY**-aware client IP handling, upstream **HTTP timeouts**, strict response validation, and **502** mapping for provider failures.
- Instrumented for operations: **Pino** JSON logs with **request IDs**, /metrics in **Prometheus** text format, optional **OpenTelemetry** export to OTLP, and Kubernetes-style **liveness** and **readiness** probes.
- Published an **OpenAPI** contract at /openapi.yaml; shipped a multi-stage, **non-root** Dockerfile with health checks, **docker-compose**, example **Kubernetes** manifests, and CI emphasizing audits, **SBOM**, and provenance-style supply-chain checks.

StreamHive | *Go, TCP/TLS, custom framing (SHV1), Prometheus, Docker*

- Authored a **Go** library and CLI experimenting with **distributed, content-addressed** blob storage over **TCP**: length-prefixed **SHV1** frames, explicit wire **handshake version**, and context-aware listen/dial with **TLS** hooks and connection limits.
- Implemented an in-memory **BlobStore** behind a stable interface, separating transport and storage so future **replication** and **peer discovery** can evolve without rewriting the framing layer.
- Exposed standard /livez, /readyz, and /metrics HTTP endpoints alongside the TCP data plane for orchestrator-friendly deployments; documented public API **semver** and architecture trade-offs in **CHANGELOG** and design docs.
- Applied mature engineering workflow: **golangci-lint**, **Makefile** targets (test, race, coverage, vet), Docker image, and CI that **pins third-party Actions to SHAs**, uploads coverage, and emits a **CycloneDX SBOM**.

Skills

Languages: Go, Python, C, PHP, JavaScript/TypeScript, SQL, Java

Frameworks: React, Vue.js, Express (Node.js), Django, Laravel, FastAPI, Symfony, Gin

ML / NLP: scikit-learn, XGBoost; text features (TF-IDF, embedding models)

Cloud / DevOps: Docker, Kubernetes, AWS, Terraform, Kafka, GitHub Actions, CI/CD, Prometheus, Grafana, OpenTelemetry

Databases & cache: PostgreSQL, MySQL, MariaDB, MongoDB, SQLite, Redis

Certifications

Algorithms Specialization (Stanford), Machine Learning (DeepLearning.AI), Meta Backend/Frontend Developer

Education

Politecnico di Torino

Sep 2025 – Present

M.Sc. in Computer Engineering

Turin, Italy

Politecnico di Torino

Sep 2021 – Sep 2025

B.Sc. in Computer Engineering

Turin, Italy

Languages

English (Bilingual), Persian (Bilingual), German (Professional), Italian (Fluent), French (Fluent)